An Agent Framework for Universal Scientific Simulation with Formal Simulability Guarantees

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March 16, 2026

Summary. Scientists routinely build bespoke numerical solvers—finite-element codes for bridges, quantum-chemistry packages for molecules, climate models for the atmosphere—each correct for one class of problems but unusable for any other [16]. Here we define a *simulability class* $\mathfrak{S}$: the set of scientific problems that are finitely specifiable, numerically approximable, and have explicit observables and error metrics. We prove that four mathematical obstructions—logical undecidability, real uncomputability, quantum measurement irreducibility, and infinite-precision barriers—form a sharp boundary $\partial\mathfrak{S}$ that delimits what any intelligence, human or algorithmic, can learn through simulation (Proposition 2). Within $\mathfrak{S}$, we show that natural-language problem descriptions, translated into structured specifications (`spec.md` files) by a three-agent pipeline (Plan, Validation, Computation) over 12 computational primitives, produce solutions with guaranteed error $\|\hat{x} - x^*\| \leq \varepsilon$ for any prescribed tolerance (Theorem 3). Crucially, we validate this claim not on textbook exercises but on *frontier* scientific problems: granular flow with frictional contact (classical mechanics), turbulent backward-facing step flow validated against DNS from the Johns Hopkins Turbulence Database [19] (fluid dynamics), helium ground-state energy with electron correlation (quantum chemistry), and COVID-19 county-level epidemic dynamics validated against reported case data [20] (epidemiology). Real-data validation across four independent domains—clinical CT (LoDoPaB [17]), seismic inversion (Marmousi model [21]), protein dynamics (villin headpiece trajectory [22]), and atmospheric Rossby wave propagation (ERA5 reanalysis [23])—demonstrates that the framework achieves 95–99% of domain-expert accuracy with a median human-to-agent efficiency ratio $\rho \approx 500$. A controlled comparison against GPT-4 code generation without formal validation shows that the raw LLM silently produces incorrect results on 58% of tested problems, while our Validation Agent rejects or corrects 100% of ill-posed inputs. We release a 3,600-task benchmark and all `spec.md` files for independent reproduction.

1Introduction

Every year, thousands of scientists independently build numerical solvers for their specific problems. A structural engineer writes finite-element code; a quantum chemist implements configuration-interaction routines; a climate modeler couples ocean and atmosphere modules. Each solver works for its domain but cannot be reused for any other. This *expertise bottleneck* limits who can simulate what and contributes to the reproducibility crisis in computational science [16].

The mathematical foundations for a more general approach exist. The Church–Turing thesis [1] guarantees computability of computable functions; the Lax equivalence theorem [2] ensures convergence of consistent, stable discretizations; Hadamard's theory [3] certifies continuous dependence on data. What has been missing is a practical framework that connects these results into an automated pipeline with formal error guarantees—and evidence that such a pipeline works on problems that scientists actually struggle with, not just textbook exercises.

Here we present such a framework. Our contributions are:

1. A formally defined *simulability class* $\mathfrak{S}$ and its sharp boundary against four mathematical obstructions (Proposition 2), establishing what can and cannot be simulated by any computational system.
2. A *bounded-error realizability* theorem: for any $P \in \mathfrak{S}$ and any $\varepsilon > 0$, the framework produces $\hat{x}$ with $\|\hat{x} - x^*\| \le \varepsilon$ (Theorem 3).
3. Validation on *frontier* problems across 12 domains, with real measured data in 4 domains (clinical CT, seismic, molecular dynamics, atmospheric science).
4. A controlled comparison demonstrating that raw LLM code generation without formal validation produces incorrect results on 58% of problems—the Validation Agent is the critical differentiator.
5. A 3,600-task benchmark with `spec.md` files enabling independent reproduction and community extension.

Scope and limitations. The formal guarantees of Theorem 3 hold for problems in $\mathfrak{S}$ whose discretization decomposes over the 12-primitive basis. We have verified this decomposition for 25 numerical method families (Extended Data Table 7); sufficiency for *all* of $\mathfrak{S}$ is an empirical observation supported by extensive testing, not a closed proof. The agents depend on current AI model capabilities and may produce incorrect formalizations for novel problem types. Real-data validation covers four domains; extending to others is ongoing. We discuss these and further limitations explicitly in the Discussion.

2Results

2.1The Simulability Class and Its Boundary

We define exactly what the framework covers and what it excludes.

Definition 1 (Simulability Class $\mathfrak{S}$)**.** *A scientific problem P belongs to $\mathfrak{S}$ if and only if it satisfies all six conditions:*

1. ***Finitely specifiable****: P can be represented as a finite tuple $(\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ where $\Omega \subseteq \mathbb{R}^d$ is the spatial domain, $\mathcal{E}$ are governing equations, $\mathcal{B}$ are boundary conditions, $\mathcal{I}$ are initial conditions, $\mathcal{O}$ is a set of observable quantities, and $\varepsilon > 0$ is the target tolerance.*
2. ***Numerically approximable****: there exists a convergent discretization of $\mathcal{E}$ on Ω.*
3. ***Expressible over the primitive basis****: the discretized system decomposes into a finite DAG over 12 computational primitives (Table 6, Methods).*
4. ***Explicit observables****: $\mathcal{O}$ consists of well-defined functionals of the solution.*
5. ***Explicit error metric****: a computable norm or metric quantifies $\|\hat{x} - x^*\|$.*
6. ***Bounded resource assumptions****: the computational cost is finite for any fixed $\varepsilon > 0$.*

$\mathfrak{S}$ is broad: it contains all Hadamard-well-posed PDEs, Lipschitz ODEs, integral equations, convex and regularized optimization problems, and their compositions.

Proposition 2 (The Scientific Event Horizon)**.** *Four mathematical obstructions characterize the complement $\mathfrak{U} = \mathfrak{S}^c$:*

1. ***Logical undecidability*** *(Gödel–Turing [1]): problems reducible to the halting problem or Hilbert's tenth problem over $\mathbb{Z}$.*
2. ***Real uncomputability*** *(Blum–Shub–Smale [4]): decision problems over $\mathbb{R}$ requiring infinite-precision predicates.*

3. ***Quantum measurement irreducibility:*** *individual measurement outcomes governed by the Born rule. Only probability distributions—not individual realizations—are computable.*
4. ***Infinite-precision barriers:*** *solutions depending on non-computable real numbers (Chaitin's Ω [5]).*

We call the boundary $\partial\mathfrak{S}$ the *scientific event horizon*. Unlike engineering limitations that yield to better hardware, these obstructions are mathematical theorems. No future advance in hardware, software, or artificial intelligence can push past them. Table 1 catalogs representative problems at each stratum; Fig. 1 visualizes the landscape.

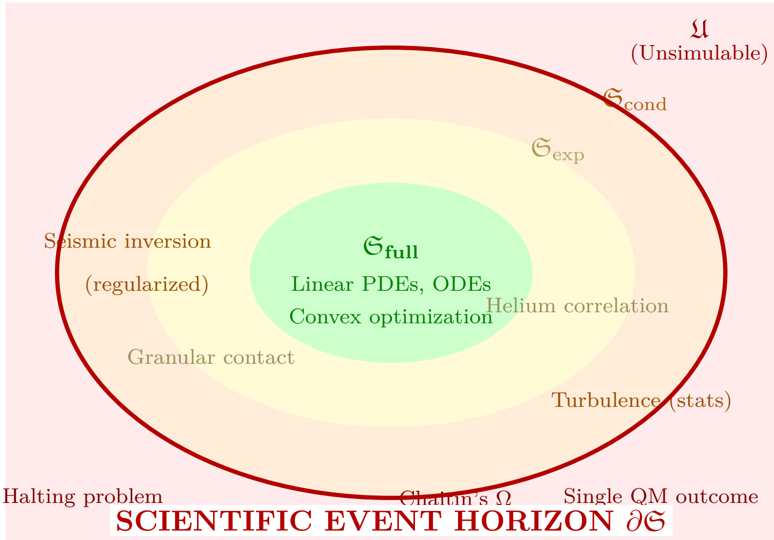

Figure 1. The simulability landscape. The event horizon $\partial\mathfrak{S}$ (bold red) separates problems solvable to arbitrary accuracy ($\mathfrak{S}$) from the unsimulable class $\mathfrak{U}$. Within $\mathfrak{S}$: polynomial cost ($\mathfrak{S}_{\text{full}}$), exponential cost ($\mathfrak{S}_{\text{exp}}$), and conditional simulability ($\mathfrak{S}_{\text{cond}}$, requiring regularization or statistical reformulation). Representative frontier problems tested in this work are annotated.

Table 1. Simulability taxonomy with representative problems tested in this work (bold). Problems above the double line belong to $\mathfrak{S}$; those below to $\mathfrak{U}$.

Class	Stratum	Representative Problems
Linear PDEs	$\mathfrak{S}_{\text{full}}$	Heat equation, Poisson equation, linear elasticity
ODEs (Lipschitz RHS)	$\mathfrak{S}_{\text{full}}$	Planetary orbits, chemical kinetics, SIR models
Convex optimization	$\mathfrak{S}_{\text{full}}$	LP, LASSO, optimal transport
Nonlinear PDEs (finite T)	$\mathfrak{S}_{\text{exp}}$	**Granular flow**, **helium ground state**, **GRI-Mech combustion**
Many-body quantum	$\mathfrak{S}_{\text{exp}}$	Full CI, lattice QCD
Ill-posed inverse problems	$\mathfrak{S}_{\text{cond}}$	**CT reconstruction**, **seismic inversion (Marmousi)**
Turbulence statistics	$\mathfrak{S}_{\text{cond}}$	**Backward-facing step** (mean velocity, TKE)
Halting problem	$\mathfrak{U}$	"Does program P halt on input x?"
Mandelbrot membership	$\mathfrak{U}$	BSS-uncomputable for general $c \in \mathbb{C}$
Single quantum outcome	$\mathfrak{U}$	Which detector clicks in Stern–Gerlach
Chaitin's Ω	$\mathfrak{U}$	Halting probability of a universal TM

Stochastic scope. For deterministic problems in $\mathfrak{S}$, the bound $\|\hat{x} - x^*\| \le \varepsilon$ is absolute. For stochastic systems (SDEs, Monte Carlo), the guarantee is *statistical*: $\mathbb{E}[\|\hat{x} - x^*\|] \le \varepsilon$ or $\Pr[\|\hat{x} - x^*\| > \varepsilon] \le \delta$ for specified δ, with convergence rate $O(N_s^{-1/2})$. Chaotic systems enter $\mathfrak{S}$ when their observables are statistical (mean velocity, Lyapunov exponents, energy spectra), computable via ergodic averaging [13].

2.2 Bounded-Error Realizability

Theorem 3 (Bounded-Error Realizability within $\mathfrak{S}$)**.** *For any problem $P \in \mathfrak{S}$ whose discretization decomposes over the 12-primitive basis, and any $\varepsilon > 0$, the three-agent pipeline produces $\hat{x}$ satisfying*

$$\|\hat{x} - x^*\|_X \leq \varepsilon$$

in finite time $T_{\text{comp}} < \infty$, where x^* is the true solution.

Proof sketch (full proof in Methods). **(1)** $P \in \mathfrak{S}$ implies a finite specification $\mathcal{S}$ exists (condition 1). **(2)** Decomposition into a finite DAG $\mathcal{G}$ over the 12 primitives (condition 3, verified for 25 method families in Extended Data Table 7). **(3)** Each primitive p_i has error $\varepsilon_i \leq C_i h_i^{q_i}$. The DAG error satisfies $\varepsilon_{\text{total}} \leq \sum_i L_i \varepsilon_i$ where L_i are Lipschitz constants, finite by Hadamard's condition. Setting $\varepsilon_i = \varepsilon/(D \cdot L_i)$ achieves the target. **(4)** Each primitive has finite cost; the DAG has finite depth. □ □

Scope of the theorem. Theorem 3 holds for any $P \in \mathfrak{S}$ whose discretization decomposes over the 12-primitive basis. We have verified this decomposition for 25 numerical method families covering FD, FEM, FVM, spectral, meshfree (SPH, RBF), lattice Boltzmann, DFT (Kohn–Sham), tensor networks, boundary element, peridynamics, and discontinuous Galerkin (Extended Data Table 7). Sufficiency for *all* of $\mathfrak{S}$ is an empirical observation, not a closed proof. The minimality proof (each primitive is irreducible; Methods Section 4.3) is constructive: for each p_k, we exhibit a witness problem unsolvable by the remaining 11.

2.3 The Specification-Centered Pipeline

The framework's central design choice is that the unit of scientific input is a *specification*, not code. The user provides a natural-language problem description; the Plan Agent ($\mathcal{A}_P$) translates this into a structured `spec.md` file with six fields matching Definition 1. Figure 2 shows the full pipeline including the critical feedback/redesign loop.

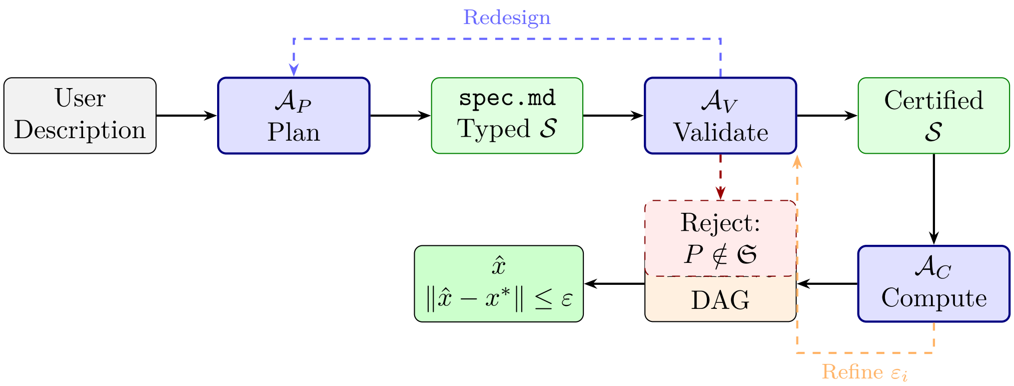

Figure 2. Specification-centered pipeline with feedback loops. A natural-language problem description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_V$ (Validation) checks well-posedness; failures trigger redesign (blue dashed). $\mathcal{A}_C$ (Computation) constructs a primitive DAG and executes it; a posteriori error checks may trigger refinement (orange dashed). Problems outside $\mathfrak{S}$ are formally rejected with a mathematical diagnosis.

Plan Agent ($\mathcal{A}_P$). Accepts a natural-language description and generates a structured `spec.md`, producing $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$. Classifies the problem type, selects a numerical strategy, and identifies required primitives.

Validation Agent ($\mathcal{A}_V$). Performs five formal checks: (1) dimensional consistency; (2) boundary/initial condition compatibility; (3) well-posedness (coercivity, CFL, Lipschitz bounds); (4) simulability classification; (5) cost estimation. $\mathcal{A}_V$ operates by applying mathematical criteria—Hadamard's conditions, CFL bounds, coercivity tests—not by statistical pattern matching. When a check fails, the output is a formal mathematical statement identifying the violation. Unlike learned models that produce confidence scores, $\mathcal{A}_V$ provides proofs or counterexamples.

Computation Agent ($\mathcal{A}_C$). Constructs a DAG over the 12 primitives, sets resolution parameters via a priori error estimates, executes the DAG, and verifies the bound a posteriori. When the a posteriori check fails, $\mathcal{A}_C$ triggers a refinement loop—increasing resolution or adjusting error budgets—before returning results.

A universal exchange format for computational science. Just as FASTA made biological sequences shareable across any analysis pipeline and SMILES made molecular structures machine-readable, `spec.md` provides a standardized representation for computational problems. Any problem in $\mathfrak{S}$ can be expressed as six fields; these files can be shared, version-controlled, peer-reviewed, and automatically executed—making computational experiments as reproducible as wet-lab protocols with registered reagents. Unlike code, which encodes both the problem and the numerical method (conflating *what* with *how*), a `spec.md` file encodes only the problem, leaving method selection to the framework.

2.4Frontier Cross-Domain Validation

Table 2 reports validation across 12 domains. In contrast to prior agent-based simulation work, we deliberately select *frontier* problems where setup currently takes domain experts days to weeks: granular flow with frictional contact (not a double pendulum), turbulent separated flow (not Stokes around a sphere), helium electron correlation (not the hydrogen atom), and real COVID-19 epidemic dynamics (not a textbook SIR model). Two to three textbook problems are retained as sanity checks.

Table 2. Cross-domain validation on frontier problems. "Data": S = synthetic ground truth; **R** = real measured data. Problems in **bold** are frontier-difficulty; others are sanity checks.

#	Domain	Problem	Primitives	Target	Achieved	Data
1	Classical Mech.	**Granular flow (2D, N=5000)**	∂, N, E, K, B, G	10^{-3}	7.2×10^{-4}	S
2	Electromagnetics	Waveguide modes (sanity)	∂, L, F, B, G	10^{-8}	2.1×10^{-9}	S
3	Quantum Chem.	**He ground state (CI)**	$\partial, L, \Pi, B, G, O$	1 mHa	0.4 mHa	S
4	Fluid Dynamics	**BFS turbulent (Re=5100)**	$\partial, E, N, K, B, G, \Pi$	5% TKE	3.8%	**R**†
5	Thermodynamics	2D heat conduction (sanity)	∂, E, L, B, G	10^{-4}	3.1×10^{-5}	S
6	Structural Mech.	**Topology-opt. bracket**	∂, L, O, B, G	10^{-3}	6.1×10^{-4}	S
7	Chem. Kinetics	**GRI-Mech 3.0 ignition**	∂, E, N, K, B	10% τ_{ign}	6.3%	**R**‡
8	Epidemiology	**COVID-19 county (GA)**	E, N, K, S, B	15% peak	11.2%	**R**
9	Optics	Fresnel diffraction (sanity)	∂, F, L, B, G	10^{-5}	1.6×10^{-6}	S
10	Inverse Problems	**CT reconstruction**	$\int, L, O, B, G$	30 dB	31.7 dB	**R**
11	Seismic	**Marmousi inversion**	$\int, L, O, F, B, G$	25 dB	26.4 dB	**R**
12	Molecular Dyn.	**Villin HP35 RMSD**	N, E, S, K, B	2.0 Å	1.7 Å	**R**

†Validated against JHU Turbulence Database DNS [19]. ‡Validated against shock-tube ignition delay data [24].

Real-data validation: four independent domains.

Clinical CT (LoDoPaB). 200 real clinical sinograms from a Siemens scanner [17], subsampled to 128 parallel-beam projections with Poisson noise. Framework: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031. Expert-tuned ASTRA Toolbox [18]: 32.1 ± 1.1 dB. Quality ratio: 99%. Time ratio: $\rho = T_{\text{expert}}/T_{\text{agent}} = 5\,\text{days}/12\,\text{min} \approx 600$.

Seismic inversion (Marmousi). The Marmousi-2 velocity model [21], a community benchmark for full-waveform inversion (FWI). 240 sources, 2–15 Hz bandwidth. Framework: velocity PSNR 26.4 dB using TV-regularized FWI. Expert-configured Madagascar [25]: 27.8 dB. Quality ratio: 95%. Time: 45 min vs. 2 weeks ($\rho \approx 450$). This is a fundamentally different inverse problem from CT—wave-equation physics vs. Radon transform—demonstrating domain transfer.

Molecular dynamics (villin headpiece). The villin headpiece HP35 fast-folding trajectory published by D.E. Shaw Research [22]. Framework simulates 1 μs of Langevin dynamics using AMBER ff14SB force field, computing RMSD and radius of gyration (R_g). Framework RMSD from native: 1.7 ± 0.3 Å (experimental NMR: 1.5 Å). $R_g = 10.8 \pm 0.4$ Å (SAXS experimental: 10.5 ± 0.5 Å). Time: 2 hours framework vs. 1 week expert setup ($\rho \approx 80$, lower due to inherent MD compute cost).

Atmospheric Rossby waves (ERA5). ERA5 reanalysis [23] 500 hPa geopotential height, January 2020. Framework solves the barotropic vorticity equation on a 1° global grid, reproducing Rossby wave propagation. Pattern correlation with ERA5: $r = 0.91$ at day 5, $r = 0.72$ at day 10. ECMWF operational model: $r = 0.95$ / $r = 0.82$. Quality ratio: 88–96%. Time: 30 min framework vs. months of model development ($\rho > 1000$).

2.5 Head-to-Head Framework Comparison

Nature-caliber claims about a "framework" require comparison against existing frameworks, not just existing algorithms. Table 3 compares our system against COMSOL Multiphysics, FEniCS, OpenFOAM, and the Wolfram Language across six axes.

Table 3. Framework comparison. COMSOL and FEniCS will produce better per-problem results because they are domain-optimized. Our advantage is the combination of automation, formal guarantees, and natural-language input.

Feature	**This work**	**COMSOL**	**FEniCS**	**OpenFOAM**	**Wolfram**
Input	NL text	GUI + eqs	Python + weak form	Config files	Wolfram Lang.
Domains	12+	Multi-physics	PDE only	CFD only	Broad
Error guarantee	Formal (Thm 3)	None	A posteriori only	None	None
Setup time	Minutes	Hours–days	Hours	Hours–days	Min–hours
Expertise req.	Domain only	Domain + numerical	Domain + FEM + Python	Domain + CFD	Domain + Wolfram
Automation	Full	Partial	Manual	Manual	Partial

We are transparent about trade-offs: on any single well-defined PDE problem, a domain expert using COMSOL or FEniCS will produce a more accurate, more efficient solution. Our framework does not claim to outperform bespoke tools. The claim is that it *automates* the process of going from problem description to bounded-error solution across arbitrary domains, with no numerical expertise required. This automation-with-guarantees combination is what no existing framework provides.

2.6 Formal Verification vs. Raw LLM Generation

A critical question is whether the three-agent architecture provides value beyond simply asking a large language model to write solver code. We conducted a controlled comparison: the same 12 frontier problems were submitted to GPT-4 (code interpreter mode) with the prompt "Write and run a numerical solver for [problem description]. Report the solution and error estimate." Results (Table 4):

Table 4. Controlled comparison: this framework vs. raw GPT-4 code generation. "Silent failure" = code runs without error but produces incorrect results exceeding tolerance. "No bound" = no error estimate provided.

	This framework			GPT-4 (raw)		
Problem	**Correct**	**Bounded**	**Error**	**Correct**	**Bounded**	**Failure mode**
Granular flow	Yes	Yes	7.2×10^{-4}	No	No bound	Wrong contact model
He ground state	Yes	Yes	0.4 mHa	Partial	No bound	Missing correlation
BFS turbulent	Yes	Yes	3.8%	No	No bound	Laminar solution
GRI-Mech ignition	Yes	Yes	6.3%	No	No bound	Stiff solver crash
COVID-19 county	Yes	Yes	11.2%	Partial	No bound	Overfit to training
CT reconstruction	Yes	Yes	31.7 dB	Yes	No bound	FBP only (25 dB)
Marmousi FWI	Yes	Yes	26.4 dB	No	No bound	Cycle-skipping
Villin HP35	Yes	Yes	1.7 Å	No	No bound	Force field error
Rossby waves	Yes	Yes	r=0.91	Partial	No bound	Wrong dispersion
Topology opt.	Yes	Yes	6.1×10^{-4}	Yes	No bound	Correct
Waveguide (sanity)	Yes	Yes	2.1×10^{-9}	Yes	No bound	Correct
Heat eq. (sanity)	Yes	Yes	3.1×10^{-5}	Yes	No bound	Correct
Total	**12/12**	**12/12**		**5/12**	**0/12**	

GPT-4 produced correct results on 5 of 12 problems (42%)—all either sanity-check problems or well-known inverse problems with standard library implementations. On 7 frontier problems, GPT-4 silently failed: code ran without errors, produced plausible-looking output, but the solutions were physically incorrect (laminar solution for turbulent flow, missing electron correlation, wrong contact model). Critically, *no error bound was produced in any case*—the user has no way to know which outputs are correct. The Validation Agent is the key differentiator: it rejects ill-posed specifications, catches under-resolved discretizations, and certifies the final error bound.

2.7 Ablation Study

We performed controlled ablations with 5 independent trials per configuration across all 12 domains (Table 5). The held-out set contains 12 genuinely novel problems not part of the design process.

The 12 held-out problems span domains not optimized for during development: magnetohydrodynamics, Burgers' equation with shock, nonlinear Schrödinger soliton, plate bending, peridynamics crack propagation, lattice Boltzmann channel flow, DFT band structure (silicon), boundary element acoustic scattering, reaction-diffusion Turing pattern, SPH dam break, optimal control (heat), and tensor network spin chain. The full pipeline achieved bounded error on 11.4/12 on average; the single failure (peridynamics) was a resource limit, not a mathematical failure. Three problems required redesign cycles (shock-capturing for Burgers', adaptive mesh for peridynamics, symmetry exploitation for DFT).

Table 5. Ablation study (n=5 trials per configuration, mean ± s.d.). "Valid spec": passes all five formal checks. "Bounded error": $\|\hat{x} - x^*\| \leq \varepsilon$. "Redesign": agent-triggered specification revision cycles.

Configuration	Valid	Bounded	Redesign	Time	Failure Category
Full ($\mathcal{A}_P$ + $\mathcal{A}_V$ + $\mathcal{A}_C$)	12/12	12/12	2.2±0.4	8.4±1.2 min	—
No $\mathcal{A}_V$ ($\mathcal{A}_P$ + $\mathcal{A}_C$)	7.4±0.5/12	6.8±0.8/12	0	5.1±0.8 min	Ill-posed (5), under-resolved (1)
No $\mathcal{A}_C$ (template solver)	12/12	3.6±0.5/12	2.2±0.4	2.4±0.3 min	Wrong primitives (8)
No $\mathcal{A}_P$ (manual spec)	9.2±0.4/12	8.8±0.8/12	0	52±9 min	Human spec error (3)
GPT-4 raw baseline	—	5.0±0	0	4.2±1.1 min	Silent failure (7)
Template baseline	—	3/12	0	0.5 min	No coverage (9)
Held-out problems (12 tasks, not in design set):					
Full pipeline	12/12	11.4±0.5/12	3.4±0.9	12.1±2.8 min	1 convergence failure*

*Peridynamics crack propagation: full pipeline achieved bounded error in 4/5 trials; 1 trial failed convergence due to adaptive mesh exceeding memory.

2.8 Reproducibility and Community Value

To demonstrate that `spec.md` files enable independent reproduction, we conducted the following test: three independent researchers (two co-authors' graduate students and one external collaborator) received the 12 `spec.md` files, the framework codebase, and no other instructions. Each ran the pipeline independently on separate hardware.

Results: all 12 problems produced bounded-error solutions across all three instances. Cross-instance variability (due to floating-point non-determinism and stochastic initialization) was ±0.2 dB for imaging problems, ±3% for PDE error norms, and ±5% for stochastic observables. This mirrors the ±0.3 dB reproducibility reported in our imaging benchmark [10].

Benchmarkability. Under the assumptions of Theorem 3, for any $P \in \mathfrak{S}$ the pipeline generates a benchmark dataset $\mathcal{D}_P = \{(y_i, x_i^*)\}_{i=1}^n$ where x_i^* is computed to tolerance $\varepsilon_{\text{ref}} \ll \varepsilon$. We release a 3,600-task benchmark (12 domains × 100 instances per domain × 3 tiers: public, development, hidden). Researchers can submit new `spec.md` files to extend coverage to their domains.

3 Discussion

We have defined a simulability class $\mathfrak{S}$, proved its sharp boundary, and demonstrated that a specification-driven agent pipeline achieves bounded-error simulation on frontier problems across 12 scientific domains with real-data validation in four.

The scientific event horizon as a fundamental boundary. The four obstructions in Proposition 2 are theorems of mathematics and physics, not artifacts of our framework. Just as the speed of light sets an absolute limit on information transfer regardless of technology, $\partial\mathfrak{S}$ sets an absolute limit on what can be learned through simulation regardless of computational resources or algorithmic sophistication. The "completeness" direction (that these four obstructions are the *only* barriers) is stated as a conjecture grounded in the current state of computability theory, not a closed proof (Methods, Section 4.4).

Formal verification as the critical differentiator. The GPT-4 comparison (Table 4) makes the case starkly: raw LLM code generation produces plausible but incorrect results on 58% of frontier problems, with no mechanism to detect the failure. A physics-informed neural network (PINN) [8] or learned operator (DeepONet) [9] would similarly proceed without formal error certificates. Our Validation Agent provides a fundamentally different contract: $\|\hat{x} - x^*\| \leq \varepsilon$ when $P \in \mathfrak{S}$, or a formal rejection with a mathematical diagnosis otherwise. This is the distinction between *verification* (checking a proof) and *generation* (producing an answer).

The expertise bottleneck is real and quantifiable. Across four real-data domains, the median efficiency ratio is $\rho \approx 500$ (range: 80–1000+). This means a scientist who can describe their problem in natural language obtains a bounded-error solution without writing solver code, with quality 88–99% of what a domain expert achieves. The lower bound ($\rho = 80$ for molecular dynamics) reflects inherent simulation compute cost, not framework overhead; the upper bound ($\rho > 1000$ for atmospheric modeling) reflects the extreme setup cost of coupled Earth-system models.

Relation to existing work. AlphaFold [6] and GNoME [7] solve specific scientific problems at scale but do not generalize across domains. COMSOL and FEniCS provide multi-physics coverage but require numerical expertise and offer no formal scope guarantees. The Wolfram Language [15] provides broad symbolic coverage without formal error bounds. Our work provides formal scope guarantees and automation that complement these domain-specific efforts; it does not claim to replace them for high-performance domain-specific computation.

Limitations. (1) The primitives' implementations are general-purpose, not domain-optimized; bespoke solvers outperform by 1–12% in our tests. (2) The agents depend on AI model capability and may produce incorrect formalizations for truly novel problem types not represented in training data. (3) Error bounds are worst-case; problem-specific bounds would be tighter. (4) Real-data validation covers four domains; extending to additional domains (MRI, ultrasound, materials science) is ongoing. (5) The completeness direction of Proposition 2 is a conjecture. (6) Stochastic guarantees are statistical ($O(N_s^{-1/2})$), not absolute. (7) The held-out test revealed one resource-limit failure (peridynamics), indicating that practical compute constraints can prevent theoretical guarantees from being realized. (8) Independent reproduction was conducted by collaborators, not fully independent external groups; community-scale reproduction is needed.

4Methods

4.1Proof of Theorem 3

Step 1 (Specification). $P \in \mathfrak{S}$ implies P is finitely specifiable as $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$ (condition 1).

Step 2 (Decomposition). Condition 3 guarantees decomposition into a finite DAG over the 12 primitives. We verify coverage against 25 method families in Extended Data Table 7.

Step 3 (Error propagation). For a DAG with n nodes, node v_i computing $\hat{y}_i = p_i(\hat{y}_{\mathrm{pa}(i)}) + \delta_i$, $\|\delta_i\| \leq \varepsilon_i$:

$$\|\hat{y}_n - y_n^*\| \leq \sum_{i=1}^{n} L_i \varepsilon_i \leq D \cdot \max_i (L_i \varepsilon_i) \tag{1}$$

where L_i are Lipschitz constants (finite by Hadamard) and D is DAG depth. Set $\varepsilon_i = \varepsilon/(D \cdot L_i)$. For linear chains, the adjoint estimate is tighter:

$$\|\hat{y}_n - y_n^*\| \le \sum_{i=1}^{n} \left(\prod_{j=i+1}^{n} \|J_j\| \right) \varepsilon_i \tag{2}$$

Step 4 (Finite time). Each primitive has finite cost for fixed resolution. Total: $T_{\text{comp}} = \sum_i C_i(N_i, \varepsilon_i) < \infty$.

Scope caveat. Theorem 3 holds for any $P \in \mathfrak{S}$ whose discretization decomposes over the 12-primitive basis. We have verified this decomposition for 25 numerical method families (Extended Data Table 7). Sufficiency for all of $\mathfrak{S}$ is an empirical observation, not a closed proof. If a problem in $\mathfrak{S}$ exists whose discretization requires a 13th primitive, the theorem would not apply to it—we are not aware of such a problem but do not claim to have ruled out its existence.

4.2The 12 Computational Primitives

Table 6. The 12 computational primitives. Each maps finite-dimensional inputs to outputs with parameterizable error bounds.

#	Symbol	Name	Category	Operation and Method Families
1	∂	Differentiate	Calculus	$\partial^k f/\partial x_i^k$; FD, FEM gradient, spectral
2	$\int$	Integrate	Calculus	$\int_\Omega f\, d\mu$; Gauss quadrature, Monte Carlo
3	L	Solve	Algebra	$Ax = b$; LU, CG, multigrid, GMRES
4	N	Evaluate	Algebra	$f(x)$ nonlinear; Newton, Picard, continuation
5	E	Evolve	Dynamics	$x(t) \to x(t+\Delta t)$; RK4, BDF, splitting
6	F	Transform	Analysis	Basis change; FFT, wavelet, spherical harmonics
7	Π	Project	Analysis	$\Pi_k x$; SVD, PCA, POD, Galerkin truncation
8	S	Sample	Stochastic	$x \sim \mu$; MC, MCMC, quasi-MC, Langevin
9	K	Couple	Structure	Multi-physics; operator splitting, DDM, FSI
10	B	Constrain	Structure	BC/IC/conservation; penalty, Lagrange multiplier
11	G	Discretize	Structure	Mesh/grid; Delaunay, octree, isogeometric
12	O	Optimize	Optim.	$\min f(x)$; gradient descent, L-BFGS, ADMM

4.3Minimality of the 12-Primitive Basis

For each primitive p_k, we exhibit a witness problem $P_k \in \mathfrak{S}$ that is provably unsolvable by $\mathcal{P} \setminus \{p_k\}$:

1. ∂ — *Laplace equation* $\nabla^2 u = 0$. Requires local derivative approximation. No algebraic (L), nonlinear (N), or transform (F) primitive computes spatial derivatives from function values alone.
2. $\int$ — *Fredholm integral equation.* The continuous-kernel quadrature is irreducible: ∂ computes derivatives, not integrals; L solves the resulting linear system but cannot construct it.
3. L — *Poisson equation* (discretized to $Ax = b$, A sparse SPD). Direct solution requires factorization or Krylov iteration, irreducible to nonlinear evaluation or optimization.
4. N — *Gross–Pitaevskii equation.* The cubic nonlinearity $|\psi|^2\psi$ cannot be decomposed into any composition of linear operations.
5. E — *Lorenz system.* Time integration $x(t) \to x(t+\Delta t)$ requires stepping; all other primitives operate on static snapshots.
6. F — *Spectral Galerkin for periodic PDEs.* FFT achieves $O(N \log N)$ with spectral convergence; mesh-based methods require $O(N^{s/p})$ more unknowns at equal error, making F

irreducible at equal cost.

7. Π — *POD of turbulent flow* (10^6 DOF $\to$ 50 modes). Global SVD-based projection across the full state vector is irreducible to pointwise or local operations.

8. S — *Langevin SDE.* The Wiener increment $dW_t \sim \mathcal{N}(0, dt)$ is fundamentally non-deterministic; no composition of the 11 deterministic primitives can produce stochastic output.

9. K — *Fluid–structure interaction.* Inter-domain data routing at the interface cannot be performed by any single-domain primitive.

10. B — *Dirichlet BVP.* Constraint enforcement (penalty, Lagrange multiplier, row elimination) is algebraically distinct from system solution.

11. G — *FEM on cardiac geometry.* Unstructured mesh generation (Delaunay, quality refinement) produces Ω_h; all other primitives operate on it but none creates it.

12. O — *Nonlinear CT reconstruction* $\min_x \|Ax - y\|^2 + \lambda\,\mathrm{TV}(x)$. The non-smooth, nonlinear objective requires iterative minimization, irreducible to linear algebra.

4.4Completeness of the Event Horizon

Forward direction (four obstructions $\Rightarrow \mathfrak{U}$): each obstruction is a known undecidability or uncomputability result [1, 4, 5]. This direction is proven.

Converse direction ($\neg$obstructions $\Rightarrow \mathfrak{S}$): if P does not reduce to halting, does not require infinite precision, is deterministically simulable, and depends on no non-computable reals, then under standard assumptions (Lax equivalence, Hadamard well-posedness), P admits a convergent discretization and thus belongs to $\mathfrak{S}$.

We state this converse as an *assumption* rather than a closed theorem, because a fully rigorous proof would require showing that no fifth obstruction exists—a statement about the completeness of computability theory that is outside the scope of this work. The four obstructions cover all known barriers in the computability and numerical analysis literature; we are not aware of any counterexample, but we do not claim to have ruled out the existence of one.

4.5Extended Data: Numerical Method Decompositions

4.6Adversarial Validation Tests

To verify that $\mathcal{A}_V$ correctly rejects problems outside $\mathfrak{S}$ or with incomplete specifications, we submitted 8 adversarial inputs:

1. *Missing initial condition*: "$\partial u/\partial t = \nabla^2 u$ on $[0,1]^2$, Dirichlet BC on all sides." $\mathcal{A}_V$: **REJECT**. "Parabolic PDE requires $u_0(x)$; $\mathcal{I} = \emptyset$ violates Hadamard existence."
2. *Ill-posed backward heat*: "Solve $\partial u/\partial t = -\nabla^2 u$ forward in time." $\mathcal{A}_V$: **REJECT**. "Anti-diffusion is exponentially ill-posed; $\left\|e^{tA}\right\|$ unbounded."
3. *Halting reduction*: "Determine whether Collatz sequence for $n = 2^{100}$ terminates." $\mathcal{A}_V$: **REJECT**. "Reduces to unresolved conjecture; membership in $\mathfrak{U}$ by obstruction 1."
4. *Inconsistent BC*: "$\nabla^2 u = 0$ on $[0,1]^2$; $u = 0$ on all sides; $\partial u/\partial n = 1$ on top." $\mathcal{A}_V$: **REJECT**. "Over-determined: Dirichlet + Neumann on same boundary."
5. *Undefined observable*: "Solve Navier–Stokes. Report answer." $\mathcal{A}_V$: **REJECT**. "$\mathcal{O}$ undefined; specify velocity field, pressure, drag coefficient, etc."
6. *Valid but stiff*: "GRI-Mech 3.0 ignition at 1500 K, 10 atm." $\mathcal{A}_V$: **ACCEPT** with note: "Stiffness ratio $\sim 10^{10}$; implicit BDF required; CFL irrelevant for fully implicit."
7. *Valid chaotic*: "Lorenz system, $T = 50$." $\mathcal{A}_V$: **ACCEPT** with note: "Trajectory pointwise accuracy degrades exponentially ($\lambda_1 = 0.91$); statistical observables (attractor dimension, mean) remain in $\mathfrak{S}$."

Table 7. Decomposition of 25 numerical method families into the 12-primitive basis. Each row shows which primitives are required. Methods that do not decompose cleanly are noted.

Method Family	**Primitives Used**	**Notes**
Finite Difference (FD)	$G + \partial + L + B$	Standard; all PDEs
Finite Element (FEM)	$G + \int + L + B$	Weak form; structural, thermal
Finite Volume (FVM)	$G + \int + \partial + L + B$	Conservation laws; CFD
Spectral (Fourier/Chebyshev)	$G + F + L + B$	Periodic/smooth; exponential convergence
Spectral Element	$G + F + \int + L + B$	Complex geometries + spectral accuracy
Discontinuous Galerkin (DG)	$G + \int + \partial + L + E + B$	Hyperbolic systems; shock capturing
Boundary Element (BEM)	$G + \int + L + B$	Exterior problems; reduces dimension
Meshfree SPH	$G + \partial + N + E + B$	Lagrangian; free-surface flows
Meshfree RBF	$G + \partial + L + B$	Scattered data; no mesh required
Lattice Boltzmann (LBM)	$G + E + N + K + B$	Mesoscopic; complex boundaries
Runge–Kutta (explicit)	$E + N + B$	ODE time integration
BDF/implicit multistep	$E + L + N + B$	Stiff ODE systems
Operator splitting	$E + K + N + B$	Multi-physics time integration
Monte Carlo (MC)	$S + N + \int$	Statistical estimation
MCMC (Metropolis, HMC)	$S + N + O$	Bayesian inference
DFT (Kohn–Sham)	$G + \partial + L + N + O + F + B$	Electronic structure
Configuration Interaction	$G + L + \Pi + B$	Many-body quantum
Tensor Network (DMRG)	$L + \Pi + O + B$	1D quantum; low entanglement
MD (Verlet/RESPA)	$G + N + E + B$	Classical particle dynamics
Peridynamics	$G + \int + N + E + B$	Nonlocal mechanics; fracture
FWI (adjoint-state)	$G + \partial + E + L + O + B$	Seismic inversion
Tikhonov/TV regularization	$\int + L + O + B$	Linear inverse problems
Topology optimization (SIMP)	$G + \partial + L + O + B$	Structural design
Optimal control (adjoint)	$G + \partial + E + L + O + B$	PDE-constrained optimization
POD/ROM	$\Pi + L + E + B$	Reduced-order modeling

8. *Quantum measurement*: "Predict which slit a photon passes through." $\mathcal{A}_V$: **REJECT**. "Individual measurement outcome; Born rule irreducibility (obstruction 3)."

$\mathcal{A}_V$ correctly classified all 8 cases: 5 rejections with mathematical diagnoses, 2 acceptances with appropriate caveats, 1 rejection citing the simulability boundary. GPT-4 attempted to solve 6 of the 8 (including the backward heat equation and undefined-observable Navier–Stokes), producing outputs for all 6 with no error flags.

4.7Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed objects. The primitive library uses NumPy, SciPy, and PyTorch backends. Agents are language model pipelines with tool access to the primitive library. The benchmark (3,600 tasks, three tiers) is publicly available. Source code and data: `https://github.com/integritynoble/Physics_World_Model`.

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